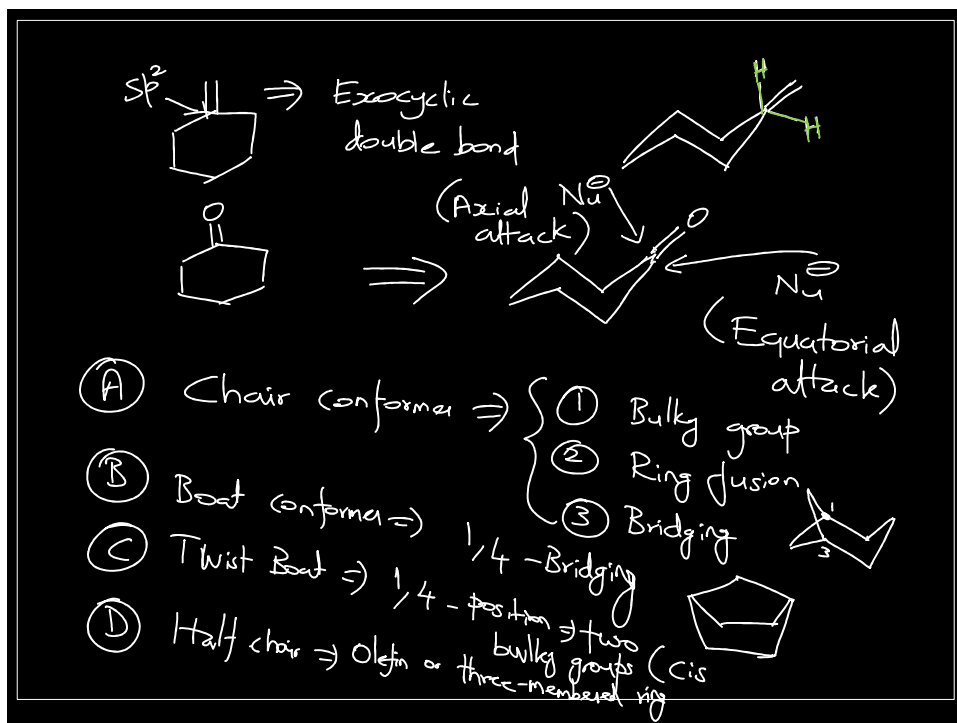
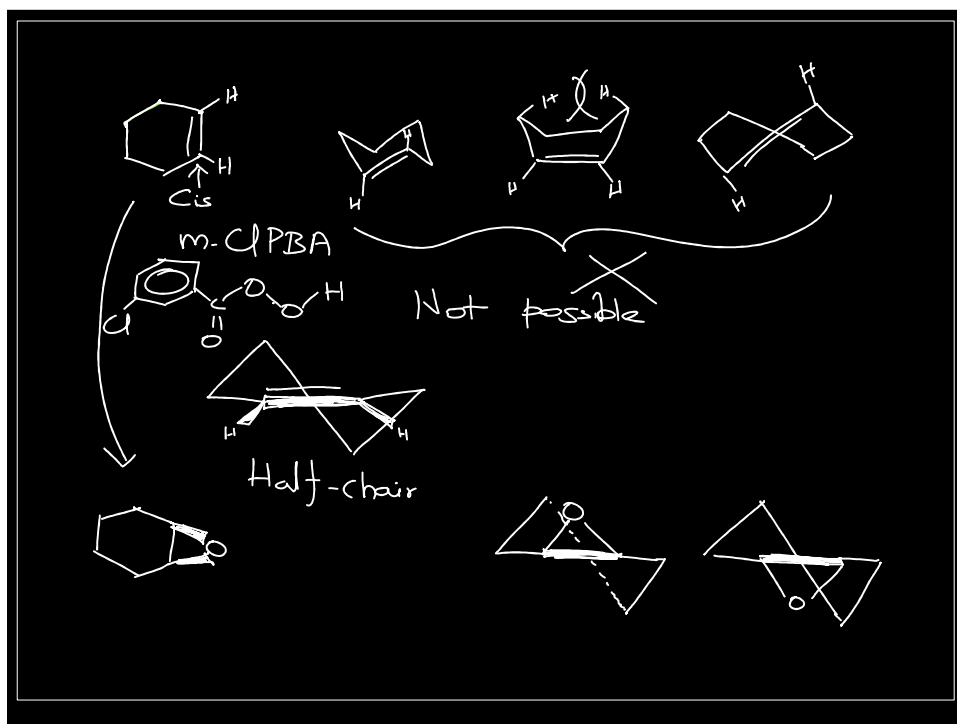
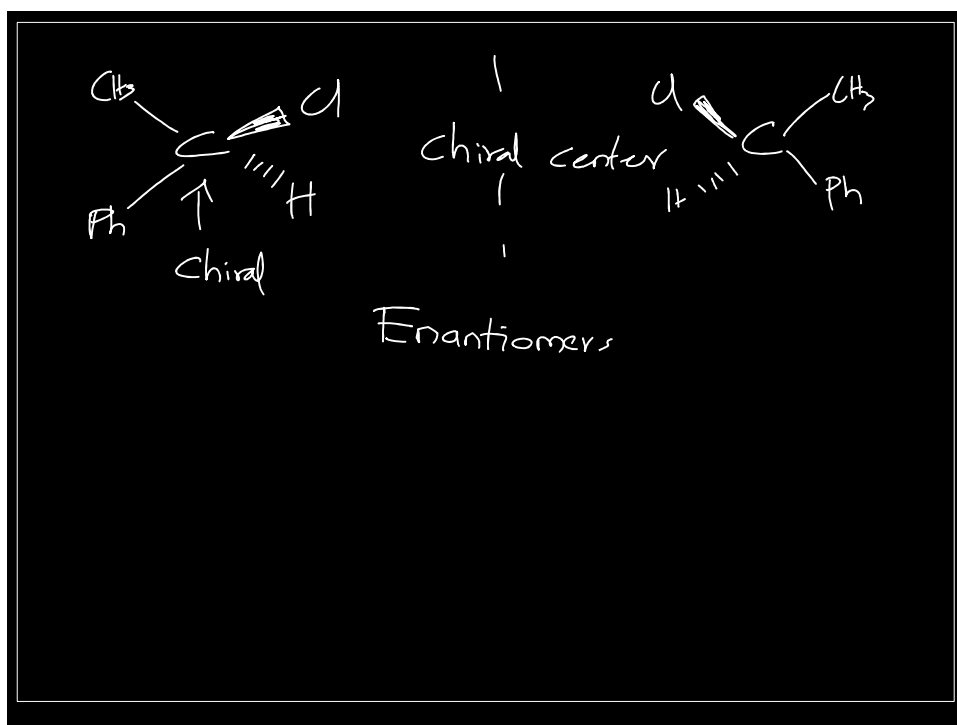
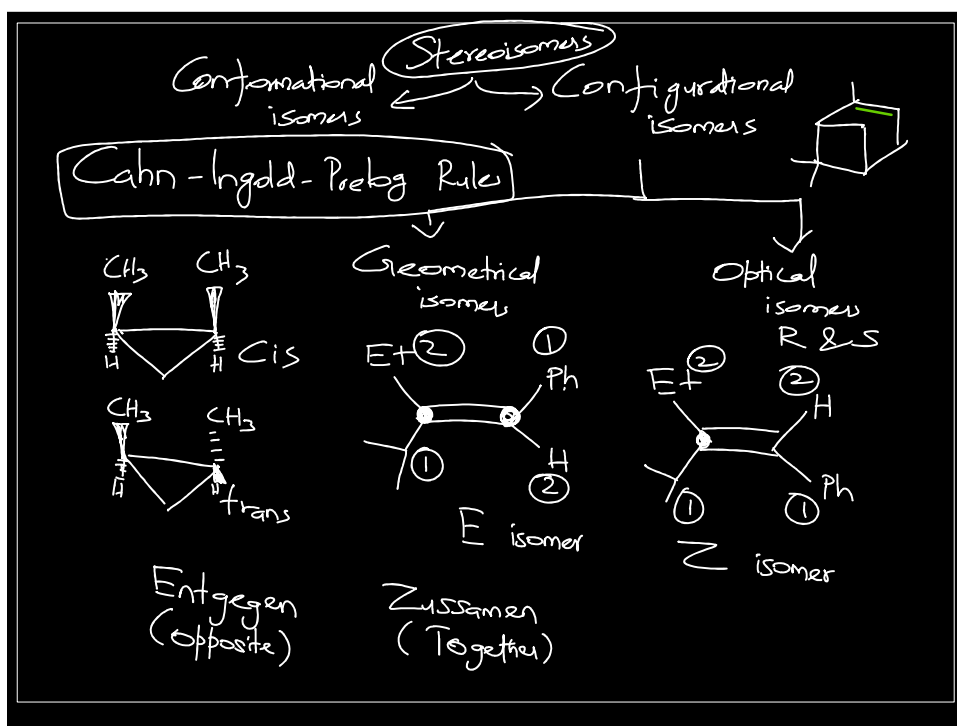
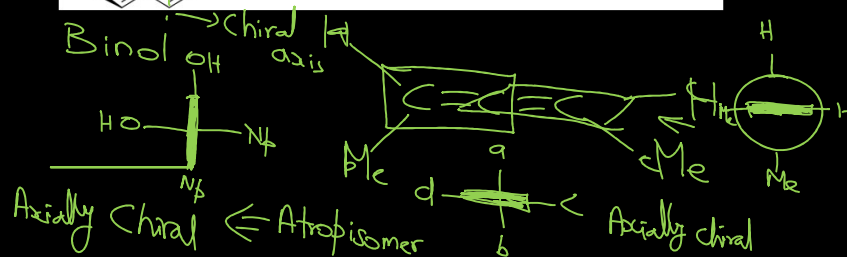
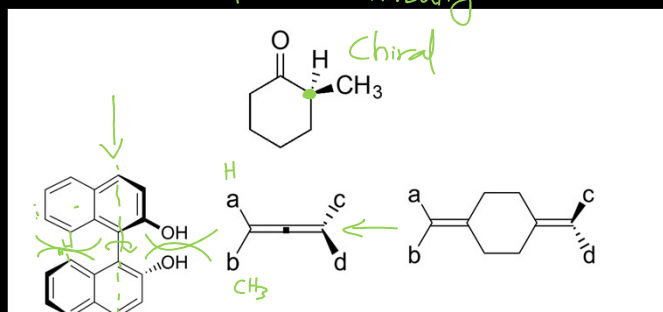




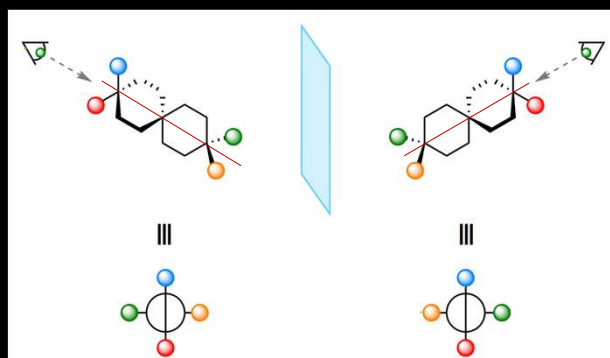
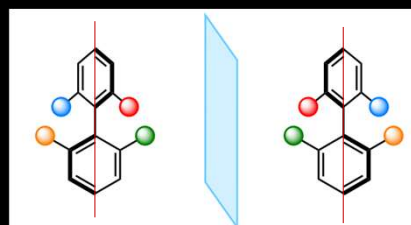
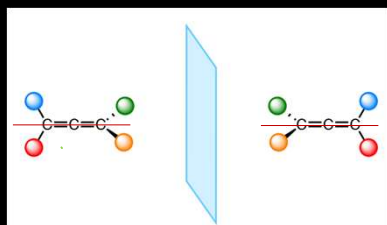


Which of the molecules are chiral?

Point chirality

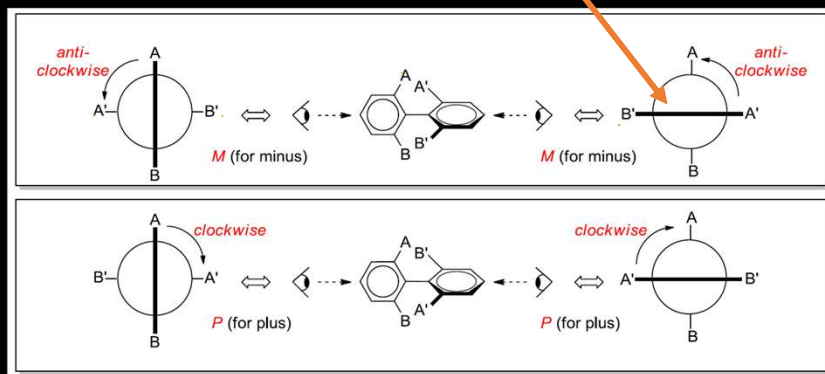


Axial chirality



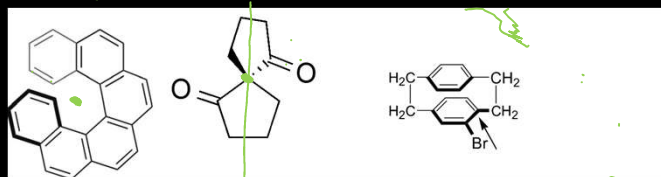
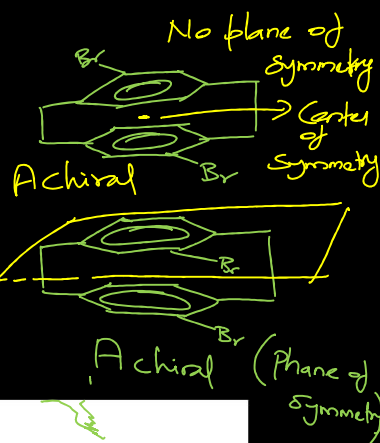
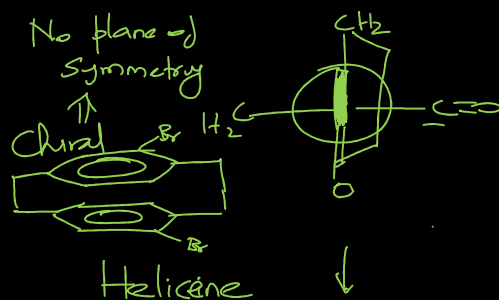
Axial chirality

Bold line indicates, the groups are near to the viewing side



Which of the molecules are chiral?

Types of chirality

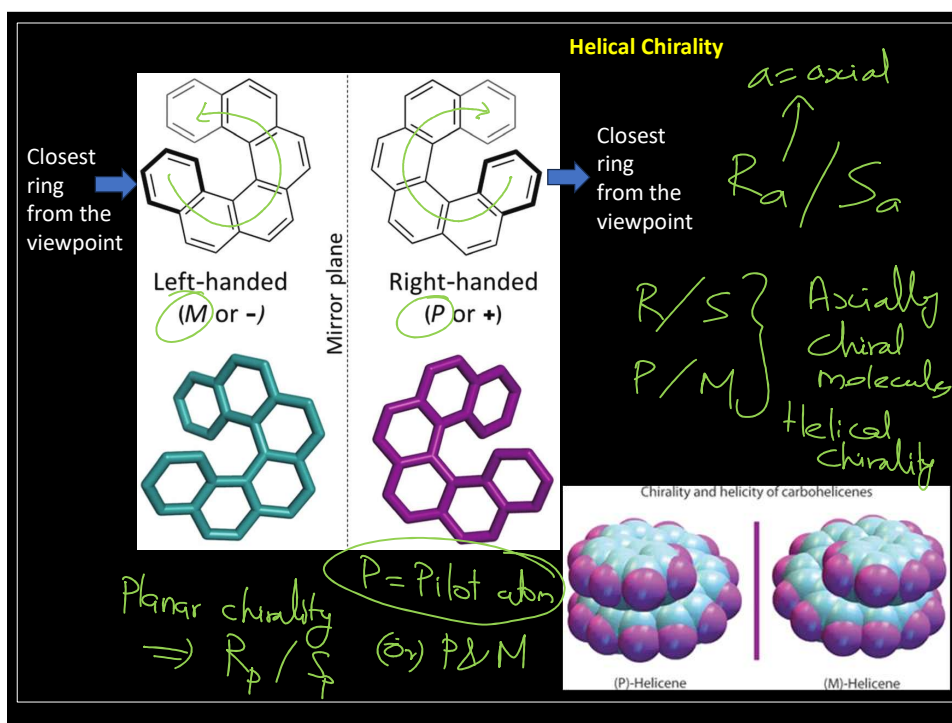


Helical chirality

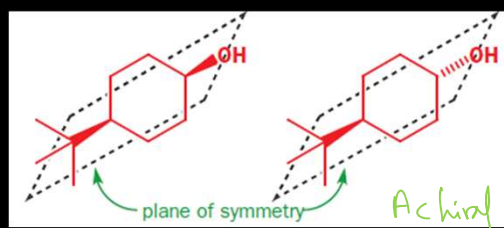
Axially chiral

Cyclophanes

Planar chirality



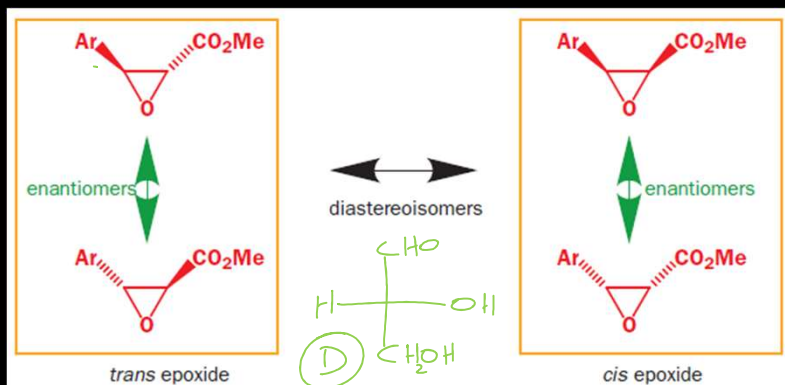
Q2. Identify whether the following molecules are chiral/achiral:



$2^n \leftarrow \text{No. of stereogenic centers}$



$$\Rightarrow 2^2 = 4$$



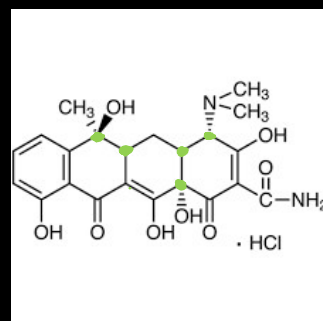
D & L Relative stereochemistry *R & S* Absolute stereochemistry *d & l / + & -*

Enantiomers are stereoisomers that are mirror images. A pair of enantiomers are mirror-image forms of the same compound and have opposite absolute stereochemistry

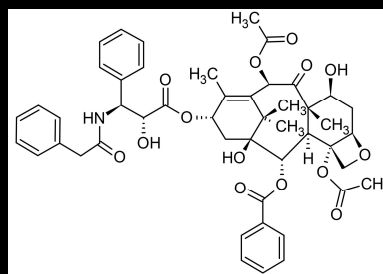
Diastereoisomers are stereoisomers that are not mirror images. Two diastereoisomers are different compounds, and have different relative stereochemistry

Q1. Identify the stereogenic centers in the following molecules:

Tetracycline is called a broad-spectrum antibiotic because it is active against a wide variety of bacteria



Paclitaxel (brand name Taxol) is a chemotherapy medication



Multi-directional Light Source

Polarizer

cell of length l (dm)

Specific Rotation = $[\alpha]$
Concentration = c
Length = l

Observed Rotation = α
 $[\alpha] = \frac{\alpha}{cl}$

$\lambda = 589 \text{ nm}$
Diastereomers
XRD
X-ray Diffraction

A. Viewing the oscillating electric fields of normal light down the axis of propagation

B. Viewing the oscillating electric fields of plane polarized light down the axis of propagation

C. Left circularly polarized + Right circularly polarized = Plane polarized

D. Faster rotating + Slower rotating = Plane is rotated

Quantification of chiral molecules: HPLC High Performance Liquid chromatography

Enantiomers $R + S$ Liquid chromatography

Diastereomers R/S R/R Identical physical properties

Different physical properties

Stationary phase: Silica gel

Adsorbate

Adsorption vs Solubility

Chemical structure of a silica gel-based stationary phase:

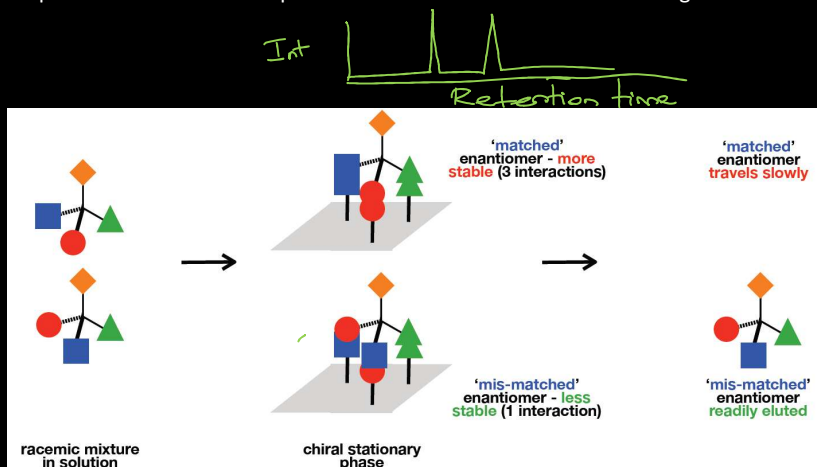
$$\begin{array}{c} \text{O-Si-O} \\ | \\ \text{O} \\ | \\ \text{Si} \end{array}$$

Chemical structure of a chiral molecule (R):

$$\text{R} = \text{CH}_3\text{CH}_2\text{CH}(\text{COOH})\text{CH}_2\text{Ph}$$

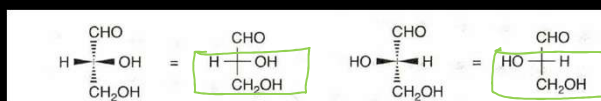
Three-point interaction

Three-point interactions and separation of enantiomers in the HPLC using chiral column



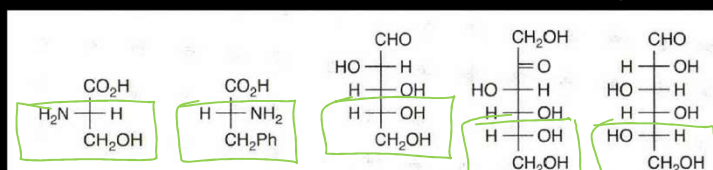
Stereochemical descriptors – Relative and Absolute stereochemistry

D/L



D-Glyceraldehyde

L-Glyceraldehyde



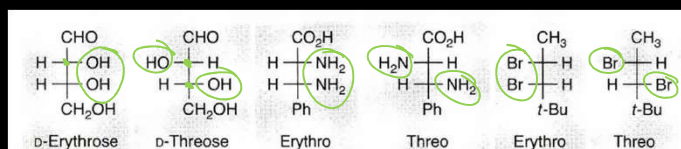
L-Serine

D-Arabinose

D

L-Idose

D-Phenylalanine

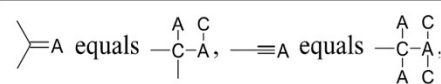
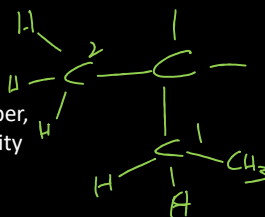


Stereochemical Descriptors

Cahn-Ingold-Prelog (CIP) conventions

The CIP convention is based on a set of sequence rules:

- (1) An atom having a higher atomic number, isotope with a higher mass has priority
- (2) If two or more of the atoms directly bonded to the asymmetric atom are identical, the atoms attached to them will be compared
- (3) Tertiary > secondary > primary.
- (4) When two groups have different substituents, the substituent bearing the highest atomic number gets priority
- (5) For multiple bonds, a doubly or triply bonded atom is duplicated or triplicated with the atom to which it is connected.



Assigning absolute configuration

